

3-long questions 24-25
1-short question marks



Statistics

Sem

2012 F

Q.1.a) The high temp.

i) here,

Total no. of data (N) = 30

Largest value (L) = 105

Smallest value (S) = 69

Then by Sturge's rule

The no. of class interval is given

by

$$K = 1 + 3.322 \log_{10} (N)$$
$$= 1 + 3.322 \log_{10} (30)$$
$$= 5.90$$

≈ 6

And the class width

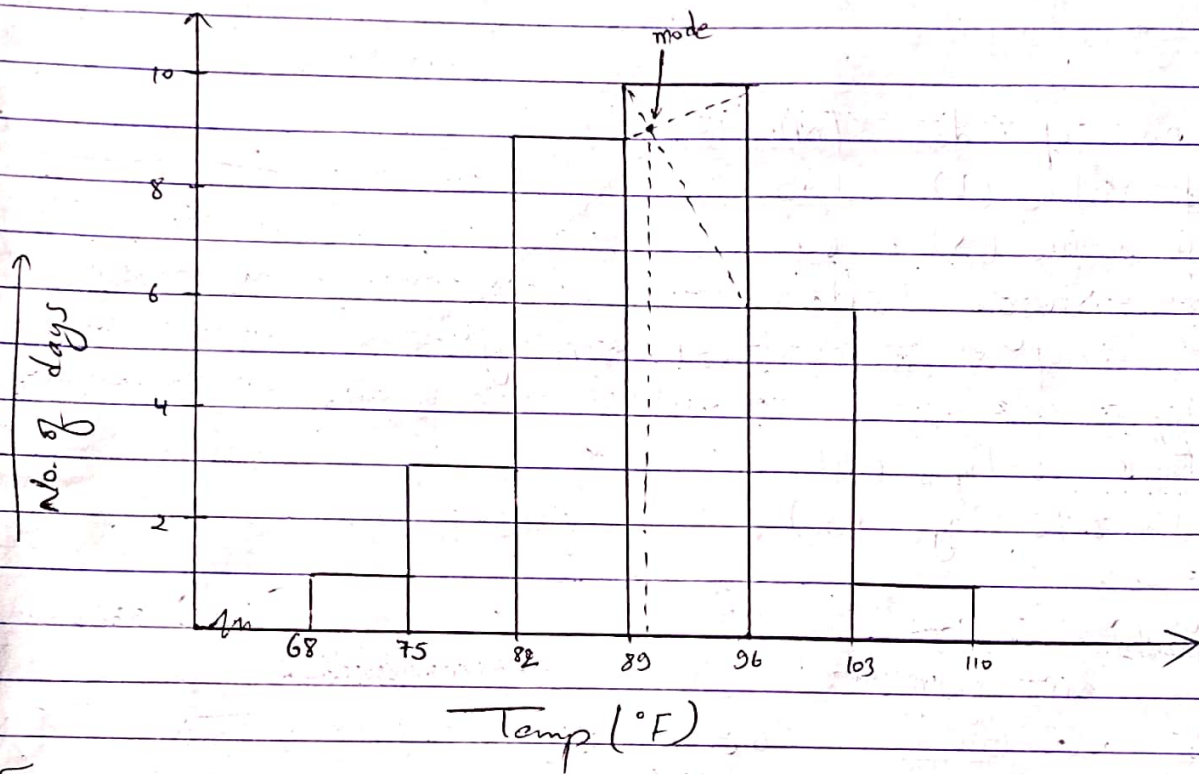
$$= \frac{L-S}{K} = \frac{105-69}{6} = 6 \approx 7$$

I would use 7 because then only there will be 6 no of class interval. If we use 6 then there will be more than 6 class which is not valid.

ii)

Temperature Class	Tally marks	frequency (No. of days)
68-75		1
75-82		3
82-89		9
89-96		10
96-103		6
103-110		1

$N=30$



* From the above histogram we can find the mode temperature is 92°F

Also, the temperature lies between 82 and 103°F most of the days.

20/2/5

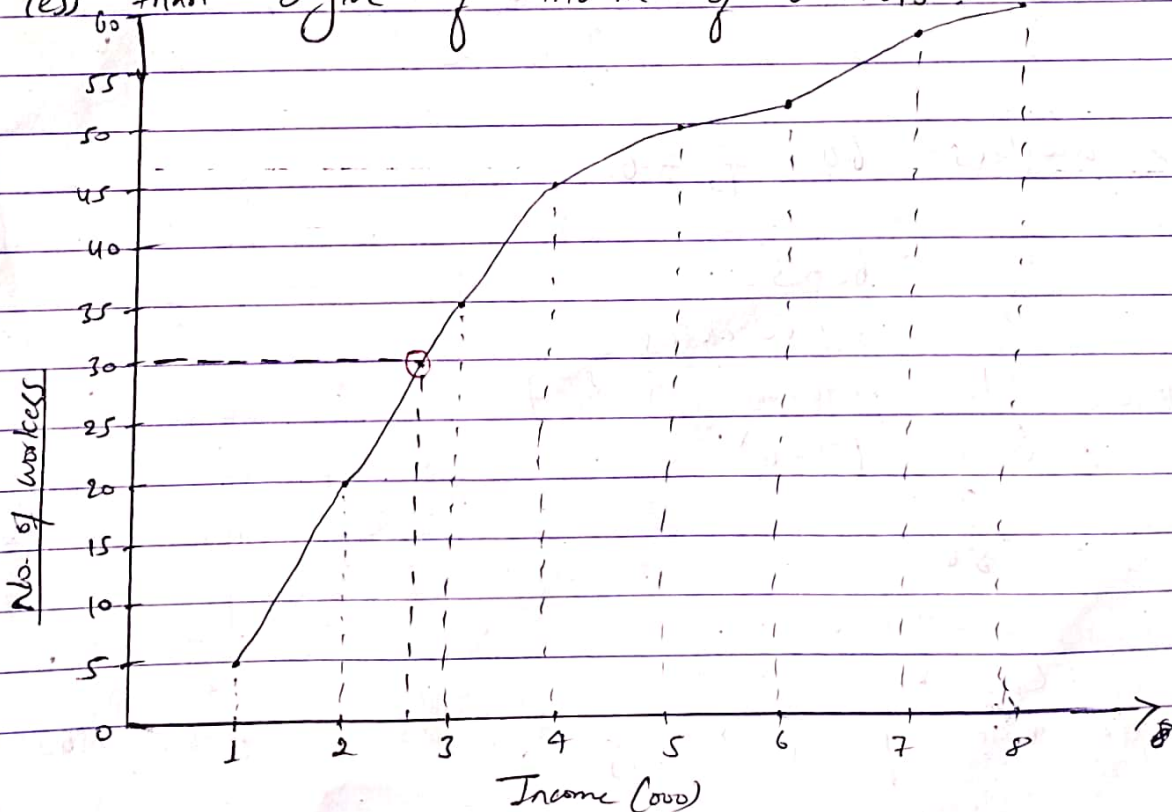
Q. Given below the pre tax
Draw less than ogive

→ Solu:-

Given frequency distribution can be written as

Income (000)	frequency	Income (000)	frequency
0-1	$60-55=5$	less than 1	5
1-2	$55-40=15$	less than 2	20
2-3	$40-25=15$	less than 3	35
3-4	$25-15=10$	less than 4	45
4-5	$15-10=5$	" " 5	50
5-6	$10-8=2$	" " 6	52
6-7	$8-2=6$	" " 7	58
7 above	2	" " 8	60

less than ogive of income of workers :-



i) Total no. of workers (N) = 60
 Lowest 50% of workers = $\frac{50}{100} \times 60$
 $= 30$ workers

For this draw a parallel line from 30 to less than of ~~give~~ give & draw a perpendicular from meeting point of O-give to the axis
 i.e. the income lies between 2 & 3
 $= 2 + \frac{2}{3} \times (3-2)$

$$= 2.67$$

$$= 2.67 \times 1000$$

$$= 2670 \text{ RS lowest}$$

Hence the highest income of the 50% workers is RS 2670

∴ ii) Top 5% workers = $60 - \frac{5}{100} \times 60$

$$= 60 - 3$$

$$= 57 \text{ workers}$$

The income lies between 6 & 7

$$= 6 + \frac{6}{7} \times (7-6)$$

$$= 6.86$$

$$= 6.86 \times 1000$$

$$= 6860 \text{ RS}$$

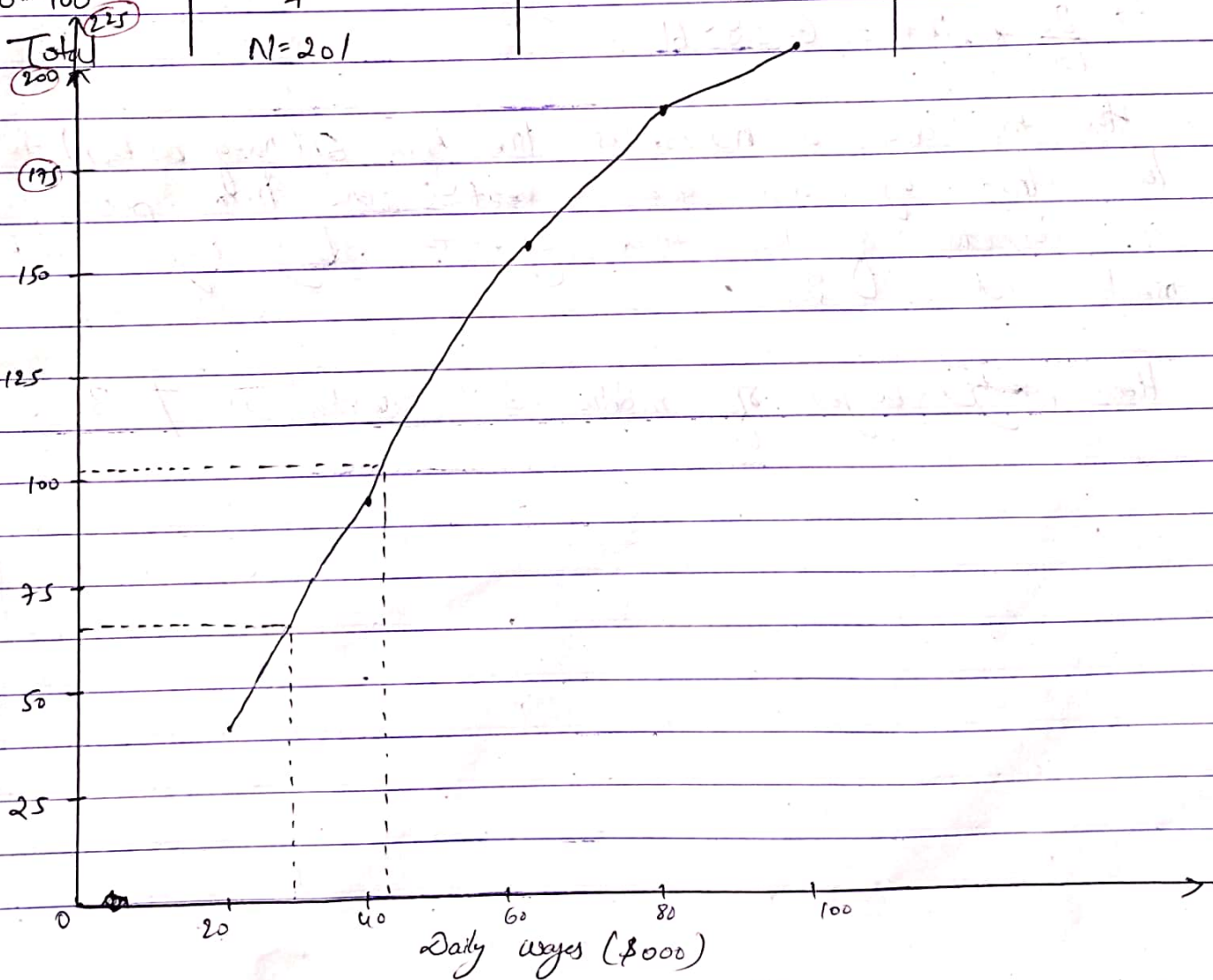
∴ Hence, the min income of top 5% of the workers is RS 6860

2018-F

1.a)

→ slu: less than frequency to distribution of daily wages is suitable for the above data.

Daily Wages (\$ 000)	No. of workers (f)	Daily wages	n/o. of workers
0-20	41	less than 20	41
20-40	51	less than 40	92
40-60	64	" " 60	156
60-80	38	" " 80	194
80-100	7	" " 100	201
<u>Total</u> (225)	N=201		



i) 50% workers $= \frac{50}{100} \times 201$
 $= 100.5 \approx 101$

for this draw a horizontal line from 101 (No. of workers) to less than ogive also draw a vertical line from point of intersection of less than ogive to daily wages which meets at 43.

Hence 50% workers earn above (more than) \$ 43000

ii) for 30%

$= \frac{30}{100} \times 241 = 60.3 \approx 61$

for this draw a horizontal line from 61 (No. of workers) to less than ogive also draw a vertical line from point of intersection of less than ogive to daily wages which meets at 30.

Hence, ~~daily~~ wages of middle 30% workers is \$ 30000

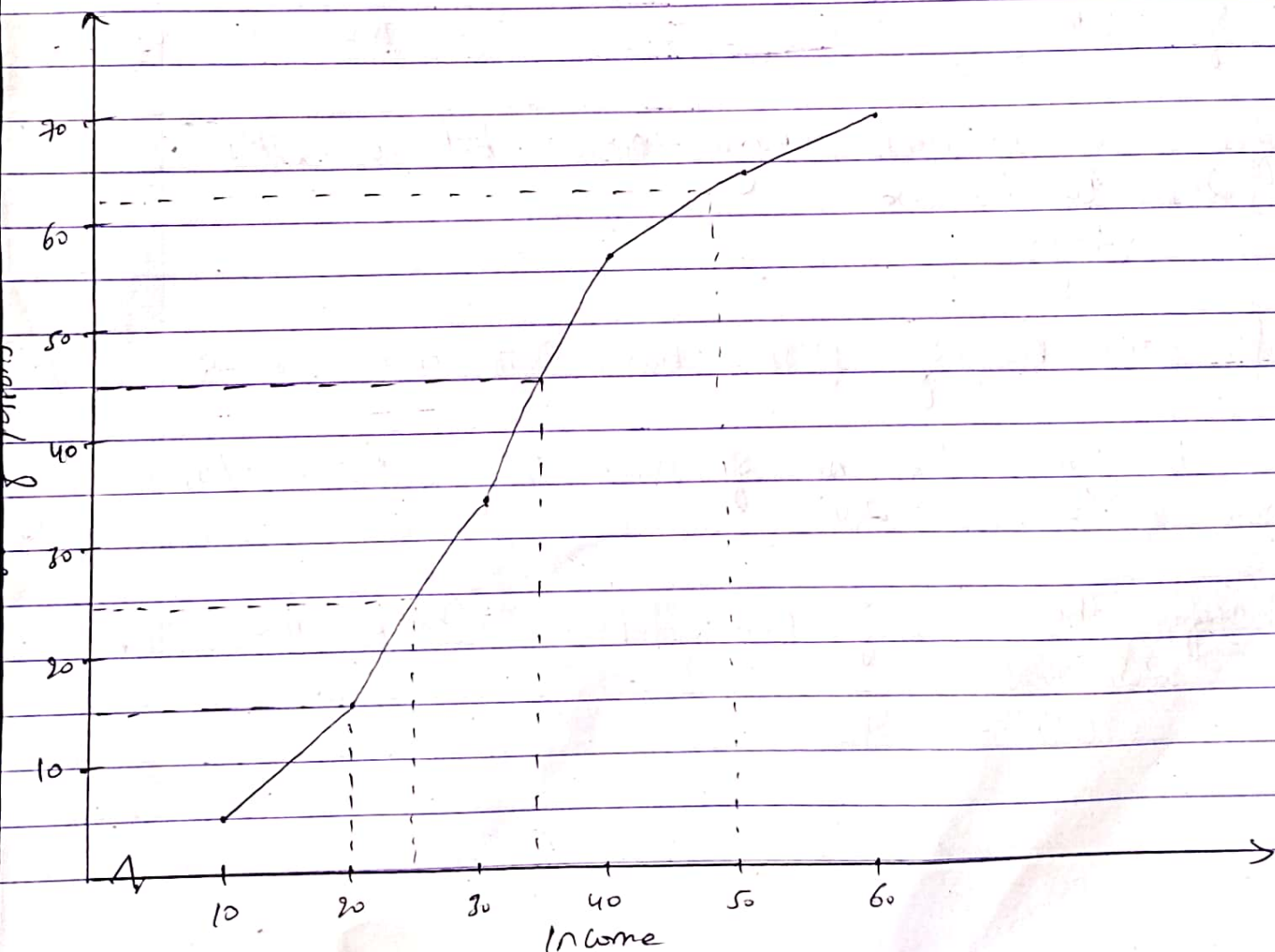
2013-F

1.6)

Less than frequency distribution of income (RS 000)

Income	no. of persons (f)	Income	no. of person
0-10	5	less than 10	5
10-20	10	less than 20	15
20-30	18	" " 30	33
30-40	23	" " 40	56
40-50	7	" " 50	63
50-60	6	" " 60	69

Total $N=69$



i) For the no. of person having income less than RS 3500
for this, Draw a vertical line from income 35 to less than give & draw a horizontal line from point of intersection to the no. of person.
It meets at 45.

Hence, 45 person have income less than RS 3500

ii) for no. of person having income between RS 20000 to 50000

From graph
No. of person having income less than RS 20000 = 15
No. of person " " " RS 50000 = 63

Therefore no. of person having income between RS 20000 to 50000
 $= 63 - 15$
 $= 48$

(iii) for the no. of person having income more than RS 25000

from the graph the no. of person having income less than RS 25000 = 24

Therefore the no. of person having income more than RS 25000
 $= 69 - 24 = 45$

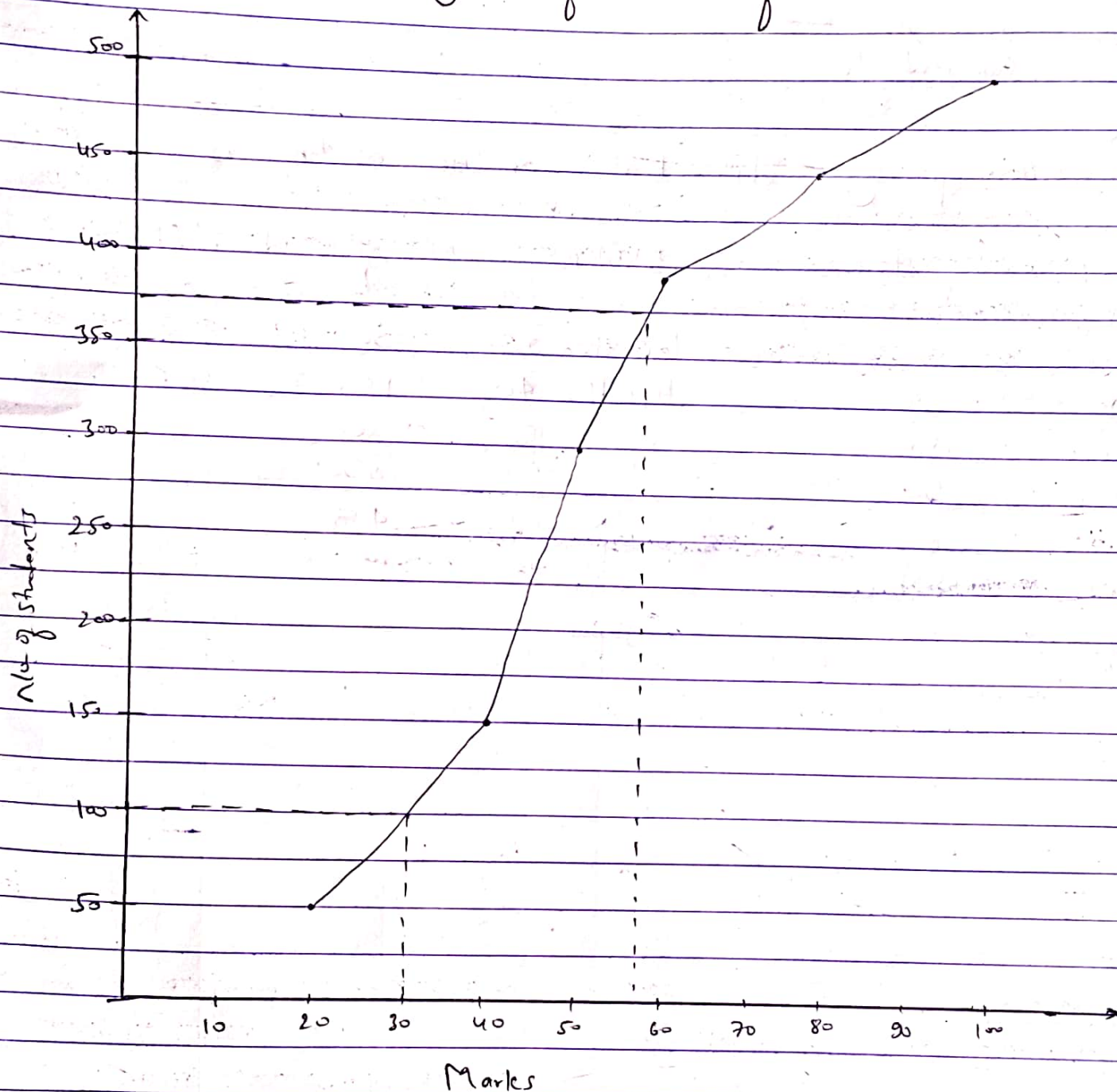
2013-5

1.6) ans. ~~The weight (in lbs)~~
for students

→ ~~sets~~ Given frequency distribution can be written as

Marks	No. of students	Marks	No. of students (f)
0-20	50	less than 20	50
20-40	100	less than 40	150
40-50	150	" "50	300
50-60	90	" "60	390
60-80	60	" "80	450
80-100	50	" "100	500

less than O-give of marks of students



Total no. of students = 500

i) Since 20% of the students are failed so,

$$\begin{aligned}\text{Number of failed students} &= 20\% \text{ of } 500 \\ &= \frac{20}{100} \times 500 \\ &= 100 \text{ students}\end{aligned}$$

for this draw a parallel line from 100 to less than O-give and draw a perpendicular line from meeting point of O-give to axis.

Hence, the highest marks obtained by 20% (100) students = 30 marks.

$$\begin{aligned}\text{ii) Top } 25\% \text{ students} &= 500 - \frac{25}{100} \times 500 \\ &= 375\end{aligned}$$

Marks lie between 50 & 60 so,

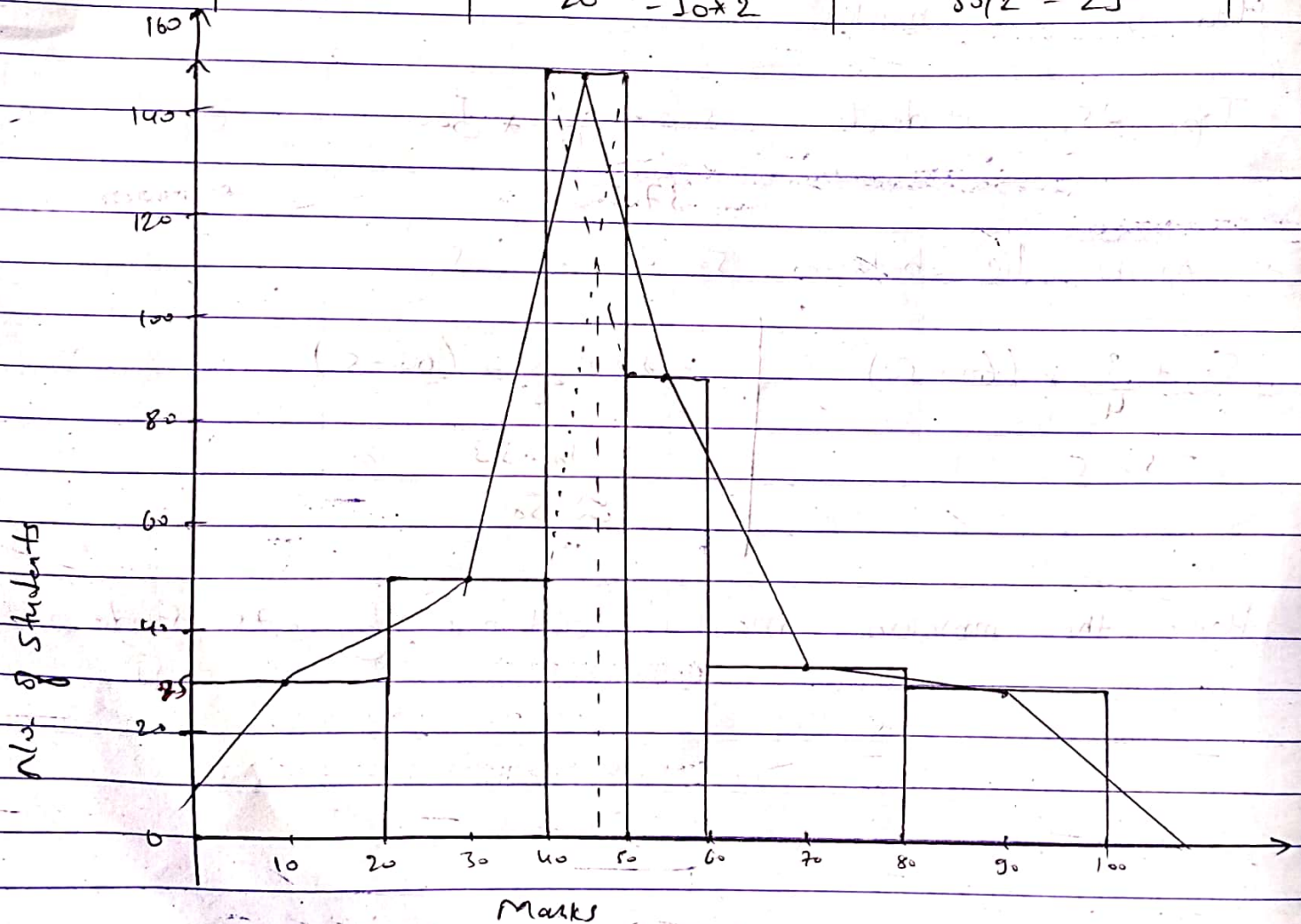
$$\begin{aligned}50 + \frac{3}{4} \times (60 - 50) \\ = 57.5\end{aligned}$$

$$\begin{aligned}&= 50 + \frac{5}{6} \times (60 - 50) \\ &= 58.33 \\ &\approx 58\end{aligned}$$

Hence the minimum marks obtained by 375 students = 58.

Also, for the histogram

Marks	No. of students	Magnitude of class	Adjusted frequency
0-20	50	$20 = 10 \times 2$	$50/2 = 25$
20-40	100	$20 = 10 \times 2$	$100/2 = 50$
40-50	150	$10 = 10 \times 1$	150
50-60	90	$10 = 10 \times 1$	90
60-70	60	$20 = 10 \times 2$	$60/2 = 30$
80-100	50	$20 = 10 \times 2$	$50/2 = 25$



Histogram & frequency polygon

Since the highest frequency is 15 which lies in the interval 40-50, mode lies between 40 to 50.

Therefore,

Modal class = 40-50

Mode = 45 (approximately)

Q. 17-F

Q. The marks obtained by 50 students

we have, total no. of data (n) = 50

$$\text{largest}(L) = 88$$

$$\text{smallest}(S) = 21$$

Then by Sturges' rule the no. of classes is given by

$$K = 1 + 3.322 \log_{10}(n)$$

$$= 6.64 \approx 7$$

$$\text{Also, the width of the class interval} = \frac{L-S}{K} = \frac{88-21}{7}$$

$$= 9.57$$

$$\approx 10$$

now frequency distribution is given by

Marks	20-30	30-40	40-50	50-60	60-70	70-80	80-90	Total
Tally marks		 	 			 		
frequency	3	7	8	11	10	6	5	50

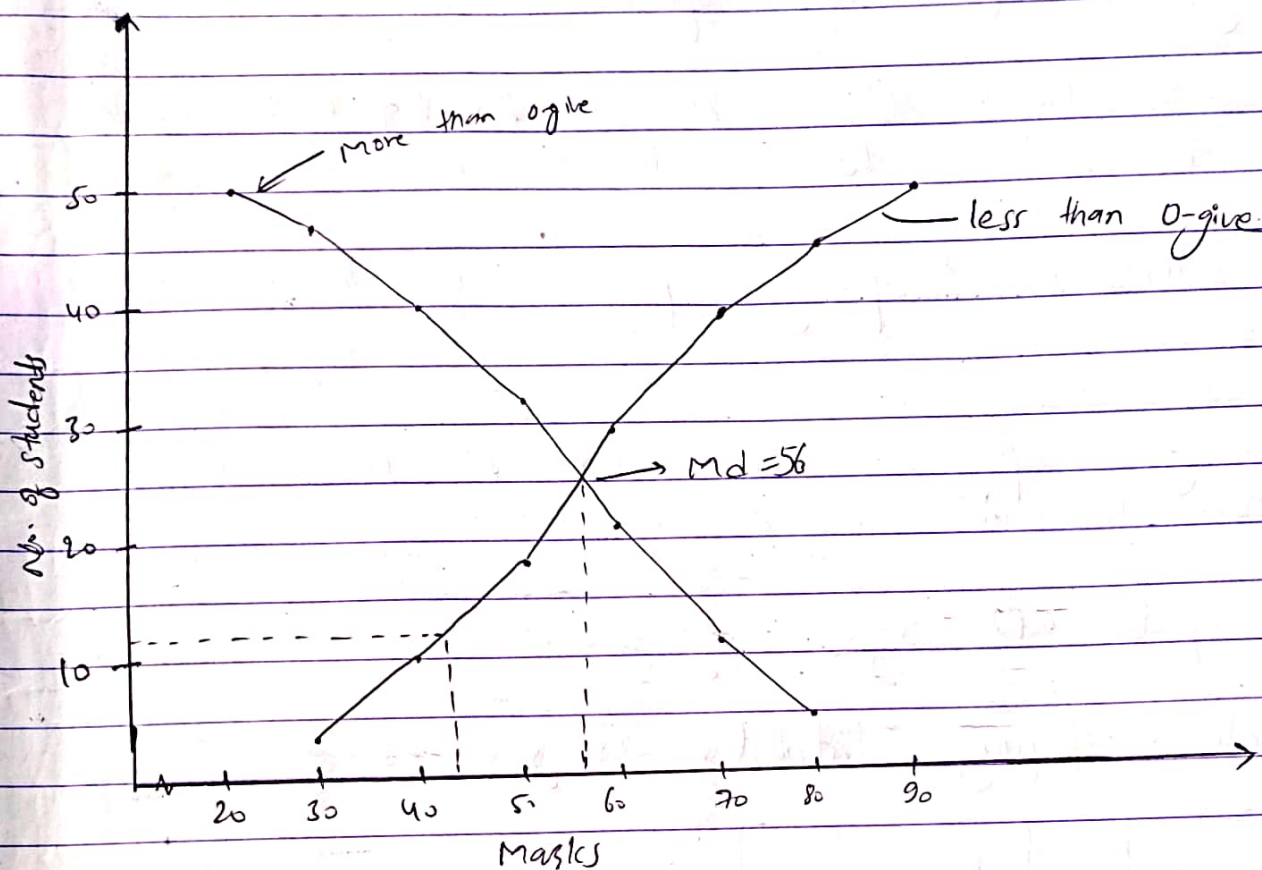
(i) from the frequency distribution table, we see that the no. of students getting 40 to 70 is more higher than getting less than 40 & more than 70 marks.

That is the no. of students getting average marks is more higher.

(ii) Cumulative frequency distribution of marks obtained by students.

Marks	Nr. of students	Marks	No. of students	Marks	Nr. of students
20-30	3	More than 20	50	less than 30	3
30-40	7	More than 30	47	" " 40	10
40-50	8	" " 40	40	" " 50	18
50-60	11	" " 50	32	" " 60	29
60-70	10	" " 60	21	" " 70	39
70-80	6	" " 70	11	" " 80	45
80-90	5	" " 80	5	" " 90	50
Total	N=50				

Cumulative frequency curve (less than O-give & more than O-give curve)



From the above graph
Median = 56

$$\therefore \text{first Quartile} = \frac{N}{4} = \frac{50}{4} = 12.5$$

from the graph first quartile = 43

$$\text{in \% first quartile} = \frac{12.5}{50} \times 100 = 25\%$$

The maximum marks obtained by first 25% student is 43.

Q: In a Survey it was found that 64 families bought milk in liter.
A frequency is given as follows:-

Milk (in litre)	5-9	10-14	15-19	20-24	25-29	30-34	35-39
frequency	7	10	13	18	8	5	3

Construct the histogram & frequency polygon of the distribution.

→ Soln:-

Since the given frequency distribution is inclusive so first we have to make it exclusive for this.

$$\text{Correction factor (CF)} = \frac{5-4}{2} = \frac{1}{2} = 0.5$$

Now, the exclusive frequency distribution is given as;

Milk (in litre)	frequency	length of class interval
4.5 - 9.5	7	5
9.5 - 14.5	10	5
14.5 - 19.5	13	5
19.5 - 24.5	18	5
24.5 - 29.5	8	5
29.5 - 34.5	5	5
34.5 - 39.5	3	5

$$N = 64$$

Since all the class interval are equal, so we don't need to adjust the frequency.

Draw histogram & polygon by same as previous question

2016-F

1.a)ans:-

Analysis sheet of pareto chart:-

Category	frequency	Percentage	Cummulative %
Control not stable	22	45.8	45.8
Operator error	13	27.1	72.9
Power fluctuation	6	12.5	85.4
Worn tool not replaced	2	4.2	89.6
Others	5	10.4	100

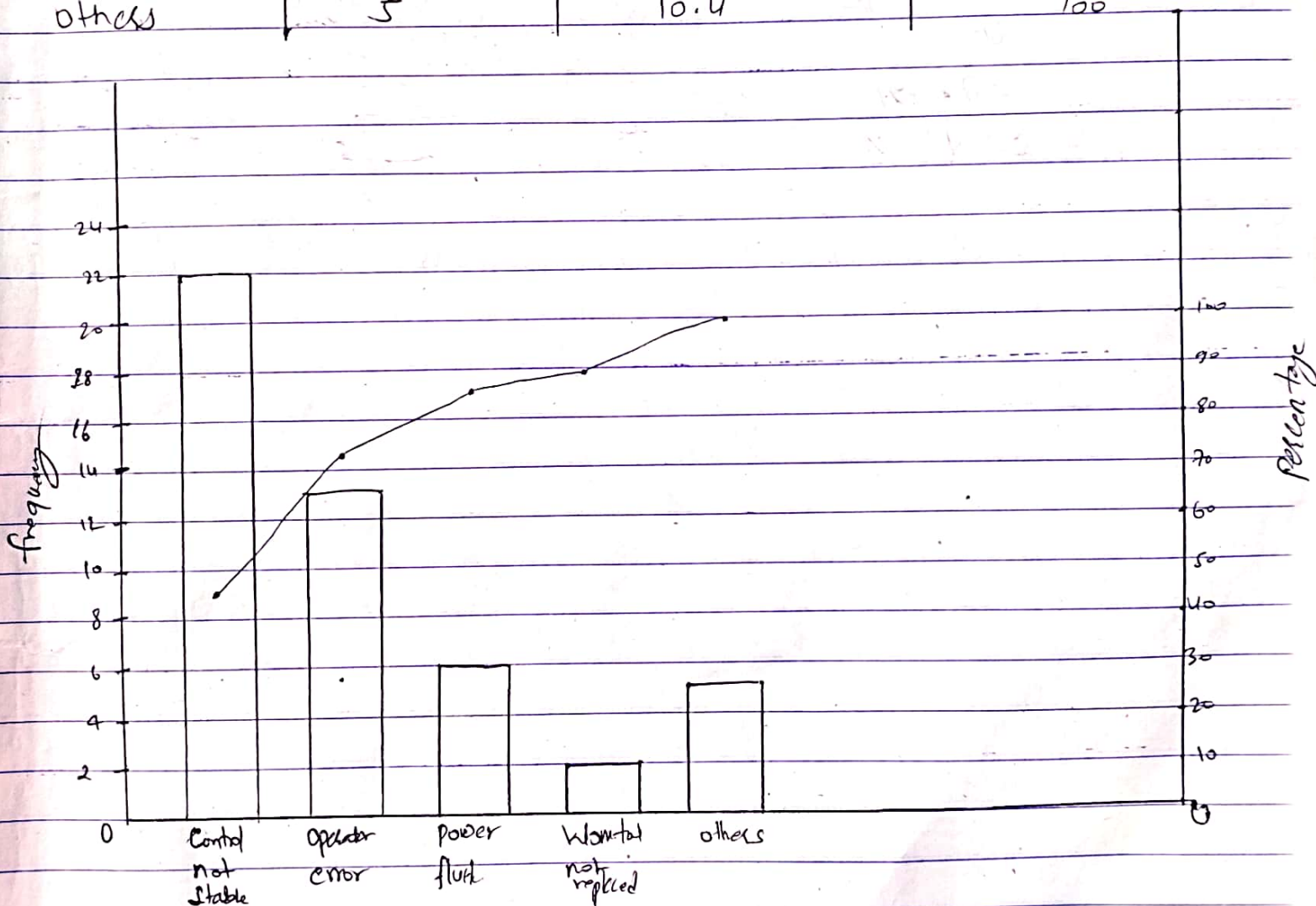


Fig. Pareto chart

i) The percentage of causes due to an unstable controller

$$= \frac{22}{48} \times 100\%$$
$$= 45.8$$
$$\approx 46\%$$

ii) The percentage of cause due to either unstable controller or operator error

$$= \frac{22+13}{48} \times 100\%$$
$$= \frac{35}{48} \times 100\%$$
$$= 72.91$$
$$\approx 73\%$$

2015-F

Solu:-

Arrange the given data in ascending order.
63, 67, 69, 71, 73, 73, 73, 75, 76, 78, 78, 81, 82, 82, 83, 83, 84, 86,
87, 88, 89, 89, 89.

i) The stem & leaf plot of marks of probability & statistics

Stem	leaf	frequency
6	3, 7, 9	3
7	1, 3, 3, 3, 5, 6, 8, 8	8
8	1, 2, 2, 3, 3, 4, 6, 7, 8, 9, 9, 9	12
9	2, 2, 4, 5, 6, 8, 9	7
		$N = 30$

ii) For the lowest score of best 25% students
For this we have to find the $(100 - 25)$
 $= 75$ percentile
i.e. P_{75}

The above table can be written as

Marks	No. of students	Cf
60-70	3	3
70-80	8	11
80-90	12	23
90-100	7	30

$N = 30$

now,

For P_{75}

$$\frac{75 \times N}{100} = \frac{75 \times 30}{100} = 22.5$$

The c.f Just greater than 22.5 is 23.

Therefore P_{75} lies in the class interval 80-90

Therefore,

$$P_{75} = l + \frac{\frac{75N}{100} - cf}{f} \times i$$

where,

$$l = 80, f = 12, cf = 11, i = 10$$

$$= 80 + \frac{22.5 - 11}{12} \times 10$$

$$= 89.58$$

Hence, the lowest score of best 25% student is 89.58

Q. The median and mean weights of 105 students of a school are 50 kg and 52 kg, respectively. Students weighting 48 kg left the school & his place was filled up by a new comer weighting 46 kg. How will the median and mean be affected by this.

→ Sol:

Total no. of students (n) = 105

Mean weight ($\bar{X}$) = 52 kg

(M_d) = 50 kg

For mean weight:-

Since the student weighing 48 kg left the school and he is replaced by the student weighing 46 kg.

The corrected mean weight of school is

$$\begin{aligned}\text{Corrected mean} &= \frac{\text{Corrected } \Sigma x}{n} \\ &= \frac{105 \times 52 - 48 + 46}{105} \\ &= 51.98\end{aligned}$$

For median weight:-

When the student weighing 48 kg is replaced by 46 kg the median remains constant because both weights are less than the median weight (52). It does not effect the position of the median.

Since the mean weight is decreased because the weight of student is replaced by less weight but the median remains constant because the no of students does not change.

2017 F | 2013 S

Q. 500 students

→ solve

Total no. of students $(N) = 500$

Since only 20% of the students had failed the exam. Then 80% of the students had passed the exam. So that the minimum pass mark is given by $(100 - 80) = 20^{\text{th}}$ percentile (i.e. P_{20})

Also, the minimum marks obtained by the top 25% of the students is given by $(100 - 25) = 75^{\text{th}}$ percentile (i.e. P_{75})

Now, Calculation of P_{20} & P_{75}

Marks	No. of students (f)	cf
0-20	50	50
20-40	100	150
40-50	150	300
50-60	90	390
60-80	60	450
80-100	50	500
$N = 500$		

For P_{20}

$$= \frac{20}{100} N = \frac{20 \times 500}{100} = 100$$

The cf just greater than 100 is 150 so P_{20} lies in the class interval 20-40

$$P_{20} = 20 + \frac{\frac{20N}{100} - cf}{f} \times i$$

here,

$$l = 20, \quad f = 100, \quad cf = 50, \quad i = 20$$

So,

$$P_2 = 20 + \frac{100 - 50}{100} \times 20 = 30$$

Similarly,

$$\text{for } P_{75} = \frac{75N}{100} = 375$$

$\therefore$ The cf just greater than 375 is 390 so, P_{75} lies on the class interval 50-60

$$P_{75} = l + \frac{75N}{100} - cf \times i$$

where,

$$l = 50, \quad f = 90, \quad cf = 300, \quad i = 10$$

So,

$$\begin{aligned} P_{75} &= 50 + \frac{375 - 300}{90} \times 10 \\ &= 58.33 \\ &\approx 58 \end{aligned}$$

Hence, the minimum ~~mark~~ pass mark is 30
& the minimum marks obtained by top 25% students is 58.

2.14

Q. J.J. Thompson - - - -

Tube 1	0.57	0.34	0.43	0.32	0.48	0.4	0.4
Tube 2	0.58	0.47	0.47	0.51	0.13	0.61	0.48

Which tube is more consistent tube?

→ Soln - Calculation of C.V for Tube 1:-

X	X^2
0.32	0.1024
0.34	0.1156
0.4	0.16
0.4	0.16
0.43	0.1849
0.48	0.2304
0.57	0.3249
$\Sigma X = 2.94$	$\Sigma X^2 = 1.2782$

$$n = 7$$

then,

$$\bar{X} = \frac{\Sigma X}{n} = \frac{2.94}{7} = 0.42$$

$$\sigma = \sqrt{\frac{\Sigma X^2}{n} - \left(\frac{\Sigma X}{n}\right)^2}$$

$$= \sqrt{\frac{1.2782}{7} - \left(\frac{2.94}{7}\right)^2} = 0.07$$

$$C.V = \frac{\sigma}{\bar{X}} \times 100\%$$

$$= \frac{0.07}{0.42} \times 100\% = 16.67$$

Similarly for Tube 2
 $C.V = 11.53$

Hence, C.V of tube 2 is less than C.V of tube 1.
 So, tube 2 is more consistent than tube 1.

2017-5

Q. Over a period of 40 days

i)

ii)

S.2 i) Arrange the given data in ascending order

60	63	64	66 66	67	71	71	71	73
74	74	76	77	78	78	78	79	79
79	79	80	80	81	81	82	82	83
83	84	85	86	86	89	90	96	98
98	99							

Stem & leaf display

Stem	leaf	frequency
6	0, 3, 4, 6, 7	5
7	1, 1, 3, 4, 4, 6, 7, 8, 8, 9, 9, 9, 9	14
8	0, 0, 0, 0, 1, 1, 2, 2, 3, 3, 4, 5, 6, 6, 9	15
9	0, 0, 6, 8, 8, 9	6
		$N = 40$

ii) for box-plot :-

Corresponding Vegetable Storage	frequency No. of days (f)	distribution cf
60-70	5	5
70-80	14	19
80-90	15	34
90-100	6	40

$$N = 40$$

smallest value (s) = 60

largest value (L) = 99

lower quartile (Q_1) :-

$$\text{Position of } Q_1 = \frac{N}{4} = \frac{40}{4} = 10$$

The c.f just greater than 10 is 19. So, Q_1 lies in 70-80 class interval.

$$Q_1 = l + \frac{\frac{N}{4} - cf}{f} \times i$$

$$= 70 + \frac{10 - 5}{14} \times 10$$

$$= 73.57$$

where, $l = 70$

$$f = 14$$

$$cf = 5$$

$$i = 10$$

Median (m_d) or Q_2 :-

$$\text{Position of } Q_2 = \frac{2N}{4} = \frac{N}{2} = 20$$

The c.f just greater than 20 is 34. So, m_d lies in class interval 80-90.

Five number summary = (S, L, Q_1, M_d, Q_3)
 $\downarrow$ smallest value $\rightarrow$ largest value

$$M_d = l + \frac{\frac{N}{2} - C.f}{f} \times i$$

$$= 80 + \frac{20 - 15}{15} \times 10 = 80.66$$

upper quartile (Q_3):-

$$\text{Position of } Q_3 = \left(\frac{3N}{4}\right)^{\text{th}} \text{ item}$$

$$= 30$$

The C.f just greater than 30 is 40, i.e. 80-90

$$Q_3 = 87.33$$

Now,

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1 = 87.33 - 73.57$$

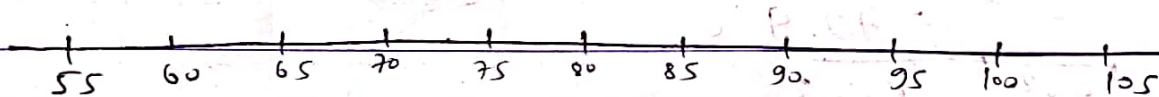
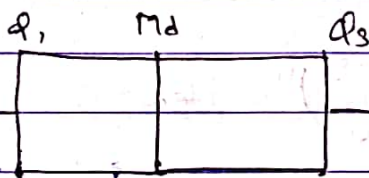
$$= 13.76$$

The whiskers & box plot is given below:-

This is also called five number summary

$$M_d - S = 20.66$$

$$L - M_d = 18.34$$



Vegetable storage

$$M_d - S > L - M_d$$

In the above figure, whisker to the left is longer than the whisker to the right. So, the distribution is negatively skewed or the distribution is left skewed.

20/7 - 5

Temp. of chemical reactor

Solu. -

Let X denotes the temperature in a chemical reactor.
Then, arranging the data in ascending order
 X (in $^{\circ}\text{C}$) = 78.1, 78.3, 78.8, 78.9, 79.2, 79.4, 80.2

Here, no. of observation (n) = 7

$$\text{i) Mean} = \frac{\sum x}{n} = \frac{78.1 + 78.3 + \dots}{7} = 78.78^{\circ}\text{C}$$

$$\text{ii) Median} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ item} = 8/2 = 4^{\text{th}} \text{ item} \\ \text{i.e. } 78.9$$

$$\text{iii) Lower Quartile } (Q_1) = \left(\frac{n+1}{4}\right)^{\text{th}} \text{ item} = 2^{\text{th}} \text{ item} = 78.3^{\circ}\text{C}$$

$$\text{iv) Upper Quartile } (Q_3) = 3\left(\frac{n+1}{4}\right)^{\text{th}} \text{ item} = 6^{\text{th}} \text{ item} = 79.4^{\circ}\text{C}$$

$$\text{v) S.d } (\sigma) = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

(Calculate the required item)

$$\text{vi) Coefficient of Variation} = C.V = \frac{\sigma}{\bar{x}} \times 100\%$$

$$= \frac{\sigma}{78.78} \times 100\% = \boxed{}\%$$

Q. type: i) which is better? $\Rightarrow$ having high mean ($\bar{x}$) makes it better
 ii) which is more consistent? $\Rightarrow$ having less C.V makes it more consistent
 iii) which is more variable? $\Rightarrow$ having more C.V makes it more variable

Q. The combined mean & variance of salary of 250 workers of Kathmandu and Bhatkapur are RS 560 and RS 597 respectively. If the mean and variance of salary of workers of Bhatkapur are RS 500 and RS 81 respectively. Find the variance salary of workers in Kathmandu.

Sol:

we have given,

Total no. of workers (N) = 250

Combined mean ($\bar{X}_{12}$) = 560

Combined Variance (σ_{12}^2) = 597

Mean salary of workers of Bhatkapur ($\bar{X}_1$) = 500

Variance " " (σ_1^2) = 81

Variance salary of worker of KTM (σ_2^2) = ?

Total no. of worker in Bhatkapur (n_1) = 150

we know,

$$N = n_1 + n_2$$

$$250 = 150 + n_2$$

$$n_2 = 100 \text{ (no. of KTM workers)}$$

now, we know combined mean is given by,

$$\bar{X}_{12} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

$$560 = \frac{150 \times 500 + 100 \times \bar{X}_2}{250}$$

$$\bar{X}_2 = 650$$

The combined variance is given by

$$\sigma_{12}^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

where,

$$d_1 = |\bar{X}_1 - \bar{X}_{12}| = |500 - 560| = 60$$

$$d_2 = |\bar{X}_2 - \bar{X}_{12}| = |650 - 560| = 90$$

Then,

$$5497 = \frac{150(81 + 60^2) + 100(62^2 + 90^2)}{250}$$

$$\sigma_{12}^2 = 121$$

$\therefore$ Hence, the variance of salary of workers in I.T.M is 121

Q. 2016-5 firm A & firm B

1.b)
ans

Sol:- Given,

Firm A

$$n_1 = 500$$

$$\bar{X}_1 = 86$$

$$\sigma_1^2 = 81$$

$$\therefore \sigma_1 = 9$$

Firm B

$$n_2 = 600$$

$$\bar{X}_2 = 175$$

$$\sigma_2^2 = 100$$

$$\therefore \sigma_2 = 10$$

For firm A

we know,

$$\text{Mean } (\bar{X}_1) = \frac{\sum X_1}{n_1}$$

$$\therefore, 186 = \frac{\sum X}{500}$$

$$\therefore, \sum X = 186 \times 500 = \text{RS } 93000$$

$$\text{C.V} = \frac{\sigma_1}{\bar{X}_1} \times 100\%$$

$$= \frac{9}{186} \times 100\% = 4.83\%$$

for firm B

we know,

$$\text{Mean } (\bar{X}_2) = \frac{\sum X_2}{n_2}$$

$$\therefore, \sum X_2 = 175 \times 600$$

$$\therefore \sum X_2 = \text{RS } 105000$$

$$\text{C.V} = \frac{\sigma_2}{\bar{X}_2} \times 100\%$$

$$= \frac{10}{175} \times 100\% \\ = 5.71\%$$

- i) Since, $\sum X_2$ is greater than $\sum X$, so, firm B has large wage bill. $\sum X_2 / \sum X = \text{total money sum}$
- ii) Since, C.V of firm B is greater than C.V of firm A. So, firm B has greater variability in wage.

iii) Combined mean :-

$$\bar{X}_{12} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

$$= \frac{500 \times 186 + 600 \times 175}{500 + 600} = 180$$

Combined variance:-

$$\sigma_{12}^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

where,

$$d_1 = |\bar{x}_1 - \bar{x}_{12}| = |186 - 180| = 6$$

$$d_2 = |\bar{x}_2 - \bar{x}_{12}| = |175 - 180| = 5$$

Then,

$$\sigma_{12}^2 = \frac{500(981 + 36) + 600(100 + 25)}{500 + 600}$$
$$= 121.36.$$

Short notes:-

I] Measures of Central tendencies:-

Central tendency is defined as the single value within the range of the data which represents a group of individual in a simple and concise manner and concentrates in the middle of the distribution.

The chief measures of Central tendencies are as follows:-

- i) Arithmetic mean (Average mean)
 - a. Simple arithmetic mean
 - b. Weighted arithmetic mean
- ii) Median
- iii) Mode
- iv) Geometric mean
- v) Harmonic mean
- vi) Percentile
- vii. Decile
- viii) Quartiles

Partition Values:-

Variate values which divides the total number of observations into an equal number of parts are called partition values. There are different types of partition values such as quartiles, deciles & percentiles etc.

1) Quartiles:-

Variate values which divides the distribution into four equal parts & each equals to 25% are called quartiles. They are denoted by Q_1, Q_2, Q_3

2) Deciles:-

- * 10 equal parts
- * 10% each
- * Denoted as $D_1, D_2, D_3, \dots, D_9$

3) Percentiles:-

- * 100 equal parts
- * 1% each
- * Denoted as $P_1, P_2, P_3, \dots, P_{99}$

- explain like quartiles

3] Application of probability & statistics in engineering: P & S

is the branch of applied mathematics & may be regarded as mathematics applied to observational data. The science of probability & statistics has been developing rapidly & its use & importance has been increasing day by day. It has various engineering applications & some of them are as follows:-

- i) It is used to test and account for the variations in materials and goods.
- ii) It is used to analyze the signal and noise.
- iii) It is used for ranging from quality control and ~~quant~~ quality assurance to communication theory.
- iv) It is used to access experimental data and control as well as improved chemical process.
- v) It is used to check the ability, performance, reliability of any system machine or equipment.

- 4] Criteria of good estimator
- 5] Errors in hypothesis testing
- 6] Coefficient of determination
- 7] Characteristics of normal distribution

4] Criteria/Characteristics of good estimator:-

- i) Unbiasedness
- ii) Consistency
- iii) Efficiency
- iv) Sufficiency

define too.

⇒ Any of numerous procedures used to calculate the value of some property of a population from observations of a sample drawn from the population is known as estimation. The criteria or characteristics of good estimator are as given below:-

8] Sources of data:-

* Definition of data - source of

* There are 2 types of data

- i) Primary data
- ii) Secondary data

i) Primary :-

data collected by an investigator directly

ii) Secondary:-

Originally collected by certain people or agency & statistically treated and the information contained in it. used again from records, processed & statistically analyzed to extract some info for other purpose.

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6.6) ans:- Solu:-

$$n = 12$$

$$\Sigma x = 132, \Sigma y = 151, \Sigma x^2 = 1944, \Sigma y^2 = 2570.98$$

$$\Sigma xy = 2233.2$$

i) the regression eqn of y on x is given by

$$y = a + bx$$

To calculate parameters a and b
we have normal equations

$$\Sigma y = na + b \Sigma x$$

f

$$\Sigma y = a \Sigma x + b \Sigma x^2$$

putting the required value on above equations

$$151 = 12a + 132b \quad \text{--- (2)}$$

$$2233.2 = 132a + 1944b \quad \text{--- (3)}$$

solving (2) & (3) eqn

$$\therefore a = -0.17$$

$$\therefore b = 1.16$$

by calculator] Solve in exam

now, putting these parameter values in eqn (1)

$$y = -0.17 + 1.16x$$

or $y = 1.16x - 0.17$ // This is the required regression equation.

ii) The correlation coefficient is given by

$$r = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$r = \frac{12 \times 2233.2 - 132 \times 151}{\sqrt{12 \times 1944 - (132)^2} \times \sqrt{12 \times 2570.48 - (15)^2}}$$

$$= 0.99$$

iii) The coefficient of determination is given by

$$r^2 = (0.99)^2 = 0.98$$

It means 98% of variation of y variables occurs due to the variation in x variable.

Also, the 98% fraction of the variance is explained by the correlation.

Rank Correlation Coefficient / Spearman Rank Correlation Coefficient

Q. Calculate the rank correlation coefficient between X and Y and also find the value of coefficient of determination and interpret the result.

X: 12 13 20 22 16 17 28 22 19 15 27 28 12

Y: 112 134 123 126 122 134 145 140 135 122 122 135 123

$$R = 1 - \frac{6 \sum d^2}{n(n^2-1)} \quad / \quad 1 - \frac{6 \left[\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \dots \right]}{n(n^2-1)}$$

→ Sol:

X	Y	R ₁	R ₂	d = R ₁ - R ₂	d ²
12	112	12.5	13	-0.5	0.25
13	134	11	5.5	5.5	30.25
20	123	6	8.5	-2.5	6.25
22	126	$\frac{4+5}{2} = 4.5$	7	-2.5	6.25
16	122	9	$\frac{10+11+12}{3} = 11$	-2	4
17	134	8	5.5	2.5	6.25
28	145	$\frac{1+2}{2} = 1.5$	1	0.5	0.25
22	140	4.5	2	2.5	6.25
19	135	7	3.5	3.5	12.25
15	122	10	11	-1	1
27	122	3	11	-8	64
28	135	1.5	3.5	-2	4
12	123	12.5	8.5	4	16

$$\sum d^2 = 157$$

here, we have arranged rank in ~~ascending~~ ^{descending} order.

$$n = 13$$

Ranks are represented so,

$$m_1 = 2, m_2 = 2, m_3 = 2, m_4 = 2, m_5 = 2, m_6 = 2, m_7 = 3$$

Then,

Rank Correlation coefficient (R) is given by

$$R = 1 - \frac{6 \left[\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \dots \right]}{n(n^2-1)}$$

here, m = No. of times that an item is repeated.

$$R = 1 - 6 \left[\frac{\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \frac{m_3(m_3^2-1)}{12} + \frac{m_4(m_4^2-1)}{12} + \frac{m_5(m_5^2-1)}{12} + \frac{m_6(m_6^2-1)}{12} + \frac{m_7(m_7^2-1)}{12}}{n(n^2-1)} \right]$$

$$= 1 - 6 \left[\frac{157 + \frac{2(4-1)}{12} + \frac{2(4-1)}{12} + \frac{2(4-1)}{12} + \frac{2(4-1)}{12} + \frac{2(4-1)}{12} + \frac{2(4-1)}{12} + \frac{3(9-1)}{12}}{13(13^2-1)} \right]$$

$$= 0.55$$

$$R = 0.55$$

Coefficient of determination

$$R^2 = (0.55)^2 = 0.3025$$

It means that there is 30.25% variance between X & Y.

Coefficient of determination of variance explain or interpret JTE 1

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Probable error of correlation coefficient :-

$$\text{Standard error } S.E.(r) = \frac{1-r^2}{\sqrt{n}}$$

$$\text{Probable error } P.E.(r) = 0.6745 \times \frac{1-r^2}{\sqrt{n}}$$

- ① If $|r| < P.E.(r)$, Correlation is not significant

② If $|r| > 6 \times P.E.(r)$, Correlation is significant

* Obtain the correct value of correlation coefficient between x & y after changing some values of x/y .
(Correcting)
(see from note copy by sir).